

25PY101: Engineering Physics

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Summative preparation

Lasers

Instructions

- Answer all questions.
- Show all the calculation steps clearly.
- Draw neat and properly labeled diagrams.
- Use SI units wherever necessary.

Questions

Q1. Numerical Problem

The wavelength of a laser is 550 nm.

- (a) Find the ratio of population of the higher energy level to the lower energy level at room temperature ($T = 300$ K).
- (b) Repeat the problem in (a) at 200 K.

Q2. Graphical / Sketch-Based Question

Sketch and label the three quantum processes of light–matter interaction:

1. Absorption
2. Spontaneous Emission
3. Stimulated Emission

Draw proper energy level diagrams for each process.

Answer

Aim:

Topic:

Given Quantities:

Working Formulas:

Steps: